Homework 1

Due data: Mar.16^{th}

Turn in your homework in class

Rules:

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism!
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit **on time**. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain **3** decimal places if rounding is required without special requirements for the question,

1 Basic Knowledge

[10 points]

(a). Please describe briefly what are ideal current source and ideal voltage source. Then, qualitatively draw their volt-ampere characteristic curves.

(b). Please briefly describe Kirchhoff's Current Law(KCL) and Kirchhoff's Voltage Law(KVL).

2 Basic Electrical Quantities

[16 points]

Consider a resistor $R = 2\,\Omega$ in the figure below (with specified direction of the voltage and current). The I–t graph shows the current flowing through the resistor. Calculate the voltage U_R. Calculate the total electric charge Q_R that flows through the resistor within 0-4s. Calculate the energy E_R absorbed by the resistor within 0-4s.

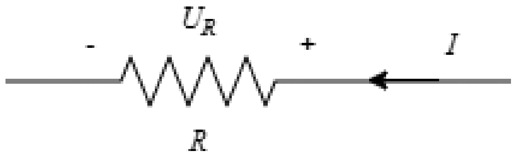

Figure 2 Resistor Diagram

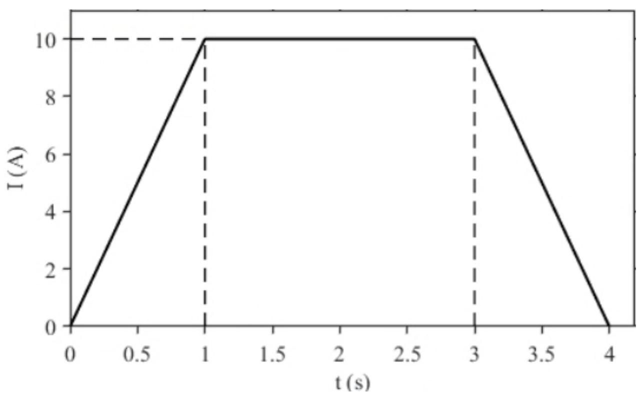

Figure 2 I-t curve

3 Wye-Delta Transformation

[12 points]The circuit is shown in **Figure 3**.
(a) Identify the type of electrical connection (either a Delta (Δ) connection or a Wye (Y) connection) formed by the nodes 2, 3, 4.
(b) Given that the current $i = 1A$, calculate the value of the resistor R. (Hint: Apply the Wye-Delta transformation to the nodes 2, 3, 4)

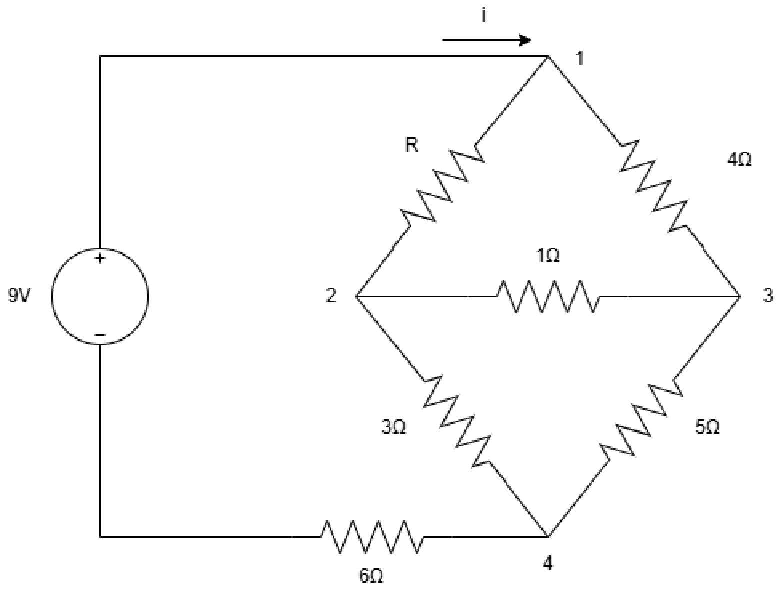

Figure 3

4 Nodal and Mesh

[14 points] The circuit is shown in **Figure 4**. Use both the **nodal analysis** method and the **mesh analysis** method to calculate V_0. Then determine whether the two results are consistent (i.e., whether they yield the same value of V_0).

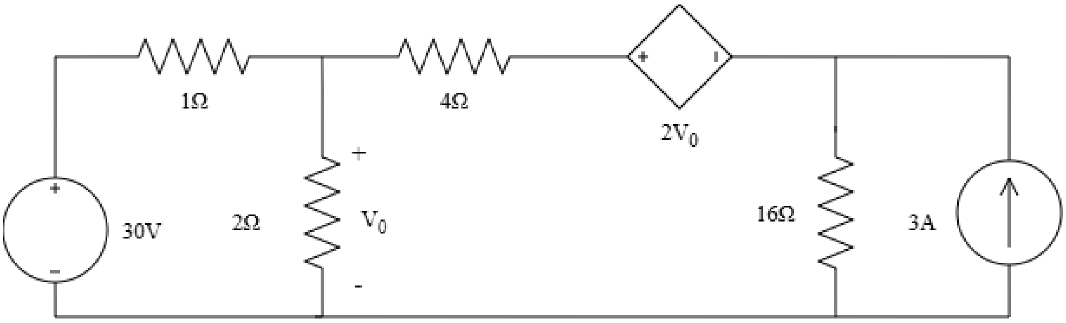

Figure 4

5 Mesh

[18 points]The circuit is shown in **Figure 5**.
(a)Define three mesh currents I_1, I_2, I_3 in the clockwise direction. Formulate the Mesh Equations (KVL) and solve for the numerical values of I_1, I_2, and I_3.
(b)Determine the actual magnitude and direction of the current flowing through the 20V voltage source. Calculate the power associated with the 20V source and specify whether it is absorbing or delivering power. Calculate the power associated with the CCVS(Current-Controlled Voltage Source) and determine whether it is consuming (absorbing) or generating (delivering) power.

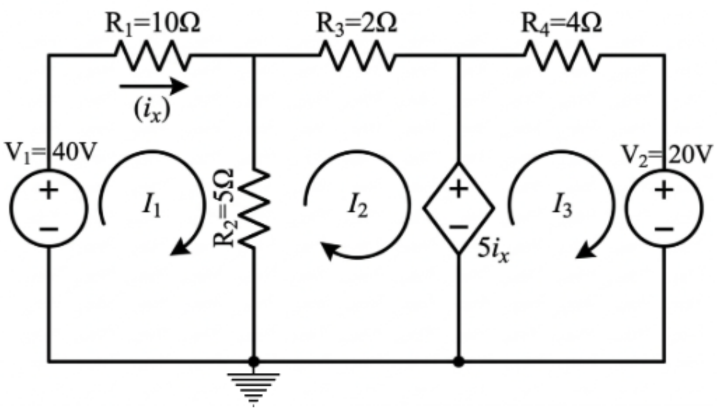

Figure 5

6 Supermesh

[14 points]Apply mesh analysis to the circuit in **Figure 6** and obtain I_0.

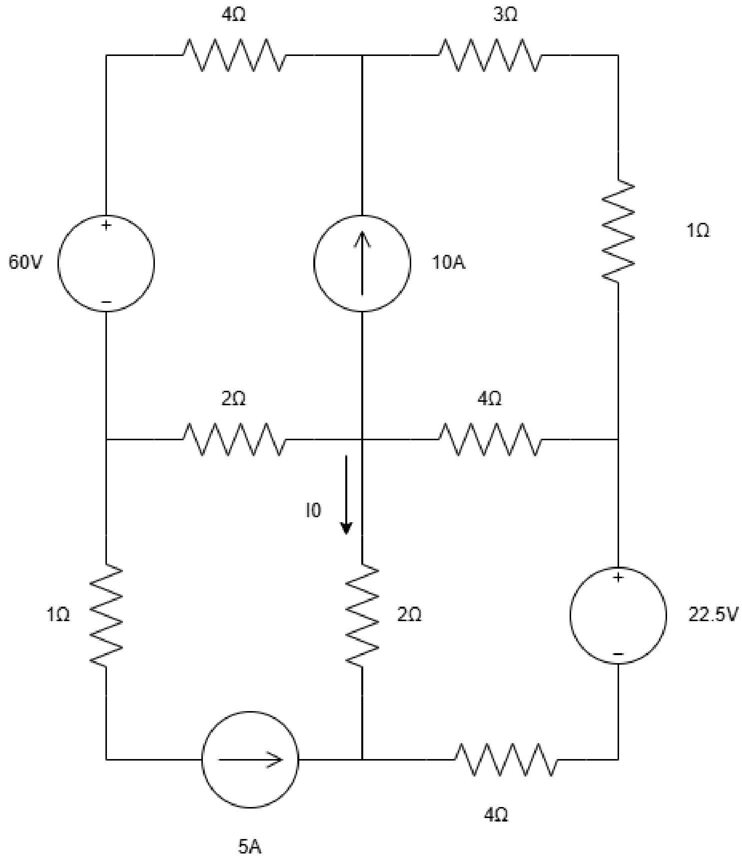

Figure 6

7 Supernode

[16 points]The circuit is shown in **Figure 7**.
$V_1 = 40V$, $V_5 = 10V$, $I_1 = 2A$. A VCCS(Voltage-Controlled Current Source) with a gain of $2V_x$, where V_x is the voltage across resistor R_1 (positive polarity on the left).$R_1 = R_2 = R_3 = 10\,\Omega$, $R_5 = 5\,\Omega$, and $R_4 = 4\,\Omega$.(Hint:Using supernode.)
(a)Calculate the node potentials V_2, V_3, and V_4.
(b)Calculate the power associated with the VCCS, P_{vccs}. State whether the controlled source is absorbing or delivering (releasing) energy based on your calculated power value.

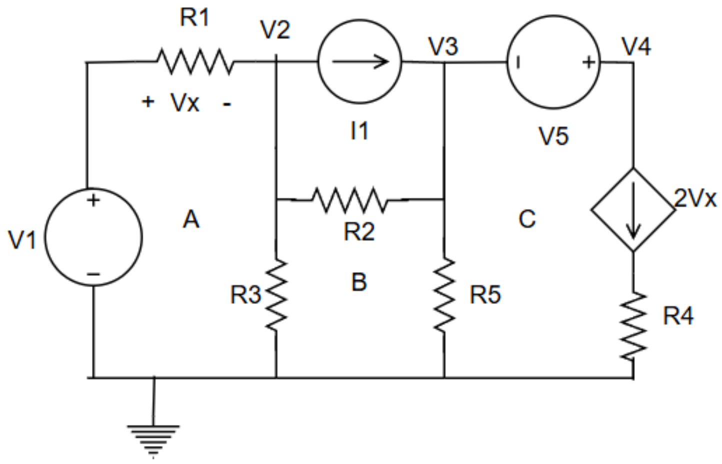

Figure 7